



Test Report No. 2024 – 0564



Agricultural Machinery – Solar-Powered Irrigation System Brgy. Candelaria, Dagohoy, Bohol

Test Report Number	2024 – 0564
Date of Issuance	November 27, 2024
Valid Until	November 27, 2029
Agricultural Machinery	Solar-Powered Irrigation System
Type	Surface Water with Reservoir, Submersible Pumpset Powered by 60 Monocrystalline Photovoltaic Modules
Pump Brand	EBARA
Pump Model	DML 622
Photovoltaic Solar Panel Brand	JINKO
Photovoltaic Solar Panel Model	JKM570N-72HL4-BDV
Test Requested by	K & C BUILDERS Block 1, Lot 29, Bridgetown Subd., Villa Kananga, Butuan City

SUMMARY

Performance Criteria	Requirement as per PNS/BAFS 324:2022	AMTEC Test
Maximum PV Performance Ratio	18 ^a	11.28
The pumpset shall have a test report with pump or performance curve prior to installation.	Provided	Not provided
Total volume discharge for whole day operation, m ³ , minimum	1300 ^b	994.09 ^c 1277.01 ^d

^a Requirement for monocrystalline PV array.

^b Requirement based on system design from 10:00 am to 3:00 pm.

^c Total volume discharge from 10:00 am to 3:00 pm.

^d Total volume discharge for whole day operation.

OBSERVATIONS

1. Operator's manual	Not provided
2. Previous test reports of similar brand and model	None
3. PV module manufacturer	JINKO SOLAR CO., LTD. No. 1, Yingbin Road, Economic Development Zone, Shangrao City, 334100 Jiangxi, P.R. China
4. Pumpset manufacturer	EBARA CORPORATION Made in China
5. Time of test	8:30 a.m. to 4:30 p.m.
6. Ease of cleaning the PV modules	The PV modules can be accessed for cleaning using a ladder or any platform.
7. Ease of adjusting and repairing of parts	All parts are accessible for adjustment and repair. However, the pump needs to be hoisted using a chain block to facilitate repairs.
8. Safety	Rotating parts, electrical wirings, and cables of the system are enclosed and properly sealed/coated.
9. Failure or abnormalities that may be observed on the system during and after the operation	No failure or abnormalities were observed on the system during and after the test operation.
10. The panels in the PV array should be installed with a rigid alloy rail made of non-corrosive material.	The PV array is installed with aluminum rails.
11. The mounting frame structure should be made from Galvanized Iron (GI) pipes or angular bars that are either primed, hot dipped galvanized with minimum of 5 mils, or double coated with non-corrosive paint.	The mounting frame structure is composed of stainless steel square tubes attached through bolts and nuts.
12. A uniform type and specifications of PV modules, either monocrystalline or polycrystalline, shall be used for the whole array.	All monocrystalline PV modules used are of uniform specifications.
13. The PV modules shall be able to withstand a minimum gustiness and uplift of 180 kph. The standard degradation should be a minimum of 0.5% annually as per IEC 61215-1:2021 (Terrestrial photovoltaic (PV) modules – Design qualification and type approval – Part 1: Test requirements).	Not verified due to limited technical capabilities on-site

OBSERVATIONS (Cont'n)

14. The orientation of the PV array shall be as per site assessment. It is recommended that the PV array should be facing the south based on efficiency and performance.	The PV array is facing south.
15. The angle of inclination in degrees for the PV array shall be dependent on the latitude of the location maximizing solar energy to be harvested.	The PV array is inclined at 22.6°.
16. Placement of the PV array shall be away from any source of shade at any time of the year. If applicable, the site of installation shall be cleared of any trees and other obstructions.	The PV array is installed in an open area, away from possible sources of shade.
17. The spatial distance between each PV module should be at least 20 mm.	The spatial distance between each PV module is 26.67 mm along its width.
18. The type, capacity of pump, suction and discharge pipe material, and other basic components shall be dependent on the design of the SPIS.	The pumpset used is submersible. An HDPE pipe was used for the discharge pipe.
19. Dry running, overheating, overloading, voltage transient, and low/high voltage input protection shall be provided for the pumpset.	Low voltage input can be detected by the controller. Dry running, overheating, overloading, voltage transient, and high voltage input protection were not verified.
20. A trash rack, made of non-corrosive material or painted with protective coating, shall be installed along with the pump intake. The orientation of the pump shall be dependent on the system design (e.g., submersible or surface).	A trash rack made of welded steel bars was provided along the pump intake.
21. The pumpset to be used shall be dependent on the water source and total head. Casings for groundwater sources shall be provided.	The system is equipped with a submersible pumpset and derives water from a nearby river.
22. A discharge measuring device, such as flow meter or water meter, should be included with the pumpset.	A discharge measuring device was not provided.
23. The pumpset controller and the PV array controller shall be insect-proofed and weather-proofed by using double-proof box and sealants.	The controller/inverter is inside a steel box.
24. A minimum of IP58 (splash proof) rating shall be used as per IEC 60529:2019 (Degrees of protection provided by enclosures [IP Code]).	The inverter has an IP20 rating as per its nameplate.

OBSERVATIONS (Cont'n)

25. The solar panel controller should be encased with a combiner box.	The controller/inverter is installed inside a combiner box.
26. The capacity of inverter shall be at least 25% higher than pumpset input power requirement.	The maximum output power of the inverter (30 kW) is 36.37% higher than the pumpset's input power requirement (22 kW).
27. The controller shall be fully automated based on manufacturers' specifications.	The controller/inverter is fully automated and has a digital display.
28. Cables and wirings used for the SPIS shall be in accordance with the Philippine Electrical Code (PEC), using PV cables specific to PV modules. Wiring installations shall be properly protected for weather conditions and other external factors.	Electrical wirings and cables of the system are enclosed and sealed/coated. Wirings and cables for the submersible pumpset are encased in a PE pipe. However, due to technical capabilities on site, it was not verified if the cables and wiring used for the SPIS are in accordance with the Philippine Electrical Code (PEC) and if the wiring installations are properly protected from weather conditions and other external factors.
29. If installed, the design of the reservoir should indicate that the intake be at least equal to the outflow during operation. The reservoir shall be able to withstand the design load. The water storage should also include a ladder, drain and overflow sensor, a freeboard, and intake and outflow valves. Construction shall be in accordance with the PD 1096.	a) A 5760-liter tank is installed in the system. The inflow valve (152 mm) is smaller than the outflow valve (203 mm). b) The reservoir was able to withstand the load when it was at full capacity. The tank is provided with a ladder, drain, and intake and outflow valves. c) It was not verified if the construction of the reservoir conformed with PD 1096 due to limited technical capabilities on site.
30. A Rapid Shutdown Device (RSD) shall be installed for roof-mounted panels.	Not applicable
31. Lightning arrester shall be installed in accordance with Article 2.80 (Surge Arresters) of the National Electrical Code (NEC).	A lightning arrester is installed above the PV array.
32. Warning signs shall be installed for electrical components on-site.	Warning notices are provided on the PV modules, and controller.
33. Surge Protection Devices (SPD) shall be installed along with the PV module.	A surge protection device is integrated into the controller.
34. Personal Protective Equipment (PPE) shall be provided for each operator on site.	The operator was not provided with personal protective equipment during the test operation.

OBSERVATIONS (Cont'n)

35. Other remarks	a. Markings and labeling are provided for the controller/inverter and PV modules. b. The test conducted conforms with the test requirement of PNS/BAFS 325:2022.
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PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing
Date of Test	November 14, 2024
Location of Test	Brgy. Candelaria, Dagohoy, Bohol 9°53'18.23" N, 124°16'59.83" E
Items Tested	Solar Panel Power Output, Input Power to the Pumpset, Pumpset Output Power, Discharge Capacity, Total Head, PV Performance Ratio, PV Module Temperature, and System Efficiency

METHODS OF TEST

The Solar-Powered Irrigation System was tested based on PNS/BAFS 325:2022 Solar-Powered Irrigation System – Methods of Test.

DESCRIPTION OF THE SYSTEM

Figure 1. Different components of the Solar-Powered Irrigation System installed at Brgy. Candelaria, Dagohoy, Bohol include the following: (a) PV modules, (b) Lightning arrester, (c) Controller box, (d) Reservoir, and (e) Discharge pipe.

DESCRIPTION OF THE SYSTEM (Cont'n)

Figure 2. Different components of the Solar-Powered Irrigation System installed at Brgy. Candelaria, Dagohoy, Bohol include the following: (a) Controller/inverter, (b) Pumpset switch, (c) Circuit breaker and Surge protection device, and (d) Intake box.

SPECIFICATIONS

Item	Manufacturer's Specification ^a	AMTEC Verification
1 Reservoir		
1.1 Overall dimensions, mm		
1.1.1 Length	ND	1700
1.1.2 Width	ND	1690
1.1.3 Height	ND	2005
1.1.4 Height from the discharge outlet of the tank to the distribution channel, mm	ND	1749
1.1.5 Thickness, mm	ND	173
2 Pumpset		
2.1 Pump		
2.1.1 Brand	EBARA	EBARA ^a
2.1.2 Model	DML 622	DML 622 ^a
2.1.3 Serial number	SP-4291	SP-4291 ^a
2.1.4 Type	Submersible, centrifugal	Submersible, centrifugal
2.1.5 Size, mm	150	150 ^a
2.1.6 Maximum discharge, m ³ /h	154.8	154.8 ^a
2.1.7 Maximum total head, m	28.3	28.3 ^a
2.1.8 Manufacturer	EBARA CORPORATION	EBARA CORPORATION ^a
2.1.9 Country of manufacture	China	China ^a
2.2 Electric motor		
2.2.1 Brand	EBARA	EBARA ^a
2.2.2 Model	DML 622_440_Three phase	DML 622_440_Three phase ^a
2.2.3 Type	Three-phase	Three-phase
2.2.4 Rated power, kW	22	22 ^a
2.2.5 Rated voltage, V	440	440 ^a
2.2.6 Rated current, A	37.6	37.6 ^a
2.2.7 Rated frequency, Hz	60	60 ^a

^a The data were obtained from the specification sheet submitted by the test applicant.

Note: ND – No Data

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
3	Controller/Inverter		
3.1	Brand	VEICHI	VEICHI
3.2	Model	S123-D5-0303-A	S123-D5-0303-A
3.3	Enclosure class	IP 20	IP 20 ^a
3.4	Input		
3.4.1	Maximum power, kW	ND	ND ^b
3.4.2	Maximum input voltage, DCV	250~780	250~780 ^a
3.4.3	Maximum input current, DCA	ND	ND ^b
3.5	Output		
3.5.1	Maximum power, kW	30	30 ^a
3.5.2	Maximum output voltage, ACV	ND	ND ^b
3.5.3	Maximum output current, A	60	60 ^a
4	Solar module		
4.1	Brand	JINKO	JINKO
4.2	Model	JKM570N-72HL4-BDV	JKM570N-72HL4-BDV
4.3	Manufacturer	JINKO SOLAR CO., LTD.	JINKO SOLAR CO., LTD.
4.4	Country of manufacture	China	China
4.5	Cell type	Monocrystalline, Bifacial	Monocrystalline, Bifacial
4.6	Cell size, L × W, mm	ND	181 × 90
4.7	Cell configuration	ND	24 × 6

^a The data were obtained from the components' nameplates.

^b Not provided on the component nameplate.

Note: ND – No Data

SPECIFICATIONS (Cont'n)

Item	Manufacturer's Specification ^a	AMTEC Verification
4.8 PV module dimension, mm		
4.8.1 Length	ND	2280
4.8.2 Width	ND	1120
4.9 Total number of solar modules	ND	60
4.9.1 Number of solar modules per string	ND	15
4.9.2 Number of strings	ND	4
4.10 Nominal power, W_p , W	570	570 ^a
4.11 Short circuit current, I_{sc} , A	$14.25 \pm 4\%$	$14.25 \pm 4\%$ ^a
4.12 Open circuit voltage, V_{oc} , V	$51.07 \pm 3\%$	$51.07 \pm 3\%$ ^a
4.13 Current at maximum power, I_{mpp} , A	13.48	13.48 ^a
4.14 Voltage at maximum power, V_{mpp} , V	42.29	42.29 ^a
4.15 Reference standard temperature condition, T_{STC}	25	25 ^a
4.16 Orientation	ND	South
4.17 Angle of inclination, °	ND	22.6
5 Safety devices	ND	Surge protection device, Lightning arrester
6 Special features	ND	Solar-powered lighting

^a The data were obtained from the component nameplate.

Note: ND – No Data

RESULTS

Table 1. Performance of the Solar-Powered Irrigation System installed in Brgy. Candelaria, Dagohoy, Bohol.

Time of the Day	Solar Irradiance, W/m ²	Solar Power, W	Module Temperature, °C	Ambient Conditions		Solar Panel Output (DC)			Input to the Pumpset (AC)			Pumpset Output			PV Performance Ratio	System Efficiency, %
				Temperature, °C	Relative Humidity, %	V	A	Power, W	V	A	Power, W	Total Head, m	Discharge Capacity, m ³ /h	Output Power, W		
8:30 a.m.	484.3	74197.4	40.9	30.5	76.8	617.9	24.5	15138.55	308.4	26.1	8049.24	13.76	117.08	4386.42	9.22	5.91
9:00 a.m.	608.5	93231.9	45.4	30.2	79.1	611.4	30.4	18586.56	333.3	30.2	10065.66	14.81	141.60	5711.47	9.26	6.13
9:30 a.m.	708.9	108614.8	46.5	32.3	4.4	610.1	33.2	20255.32	340.4	31.6	10756.64	15.20	150.19	6216.51	10.10	5.72
10:00 a.m.	745.3	114197.0	44.3	31.1	73.0	616.0	38.6	23777.60	366.6	36.9	13527.54	17.95	194.71	9517.66	8.44	8.33
10:30 a.m.	990.8	151806.4	41.2	32.9	74.7	680.0	38.8	26384.00	375.0	39.9	14962.50	17.87	193.38	9409.17	10.15	6.20
11:00 a.m.	1038.2	159068.9	44.9	33.0	72.9	605.0	38.2	23111.00	374.9	38.5	14433.65	17.97	194.71	9530.14	11.02	5.99
11:30 a.m.	824.0	126255.1	49.0	34.1	68.0	602.5	40.0	24100.00	367.7	37.1	13641.67	18.19	197.89	9805.18	9.26	7.77
12:00 m.	928.5	142255.9	52.8	35.2	64.1	646.7	40.5	26191.35	375.2	38.6	14482.72	17.95	194.42	9505.24	9.82	6.68
12:30 p.m.	844.4	129370.5	47.4	34.4	63.0	606.7	41.3	25056.71	370.5	37.2	13782.60	18.84	203.09	10422.05	9.39	8.06
1:00 p.m.	899.1	137761.6	45.9	34.6	63.0	631.5	41.5	26207.25	375.2	38.5	14445.20	21.31	202.45	11750.22	9.54	8.53
1:30 p.m.	571.7	87588.5	47.2	34.7	63.1	608.4	28.5	17339.40	320.5	27.6	8845.80	15.57	157.43	6674.85	9.90	7.62
2:00 p.m.	593.1	90867.3	44.7	32.5	64.2	607.2	29.2	17730.24	327.2	28.5	9325.20	15.75	157.33	6748.78	9.74	7.43
2:30 p.m.	611.1	93625.2	47.2	34.1	63.6	598.6	28.5	17060.10	320.8	27.6	8854.08	15.42	154.29	6478.25	10.57	6.92
3:00 p.m.	425.9	65254.7	39.2	31.8	66.5	635.3	17.7	11244.81	282.4	21.0	5930.40	14.05	126.74	4849.78	11.00	7.43
3:30 p.m.	354.2	54274.2	38.5	32.6	65.0	620.9	15.8	9810.22	263.2	18.7	4921.84	13.05	102.44	3639.75	11.03	6.71
4:00 p.m.	198.8	30465.0	34.4	32.0	69.7	619.6	9.9	6134.04	221.7	12.6	2793.42	11.48	47.03	1469.85	10.91	4.82
4:30 p.m.	166.4	25491.6	33.5	31.3	71.6	554.2	8.7	4821.54	205.5	11.0	2260.50	11.11	19.23	581.94	11.28	2.28

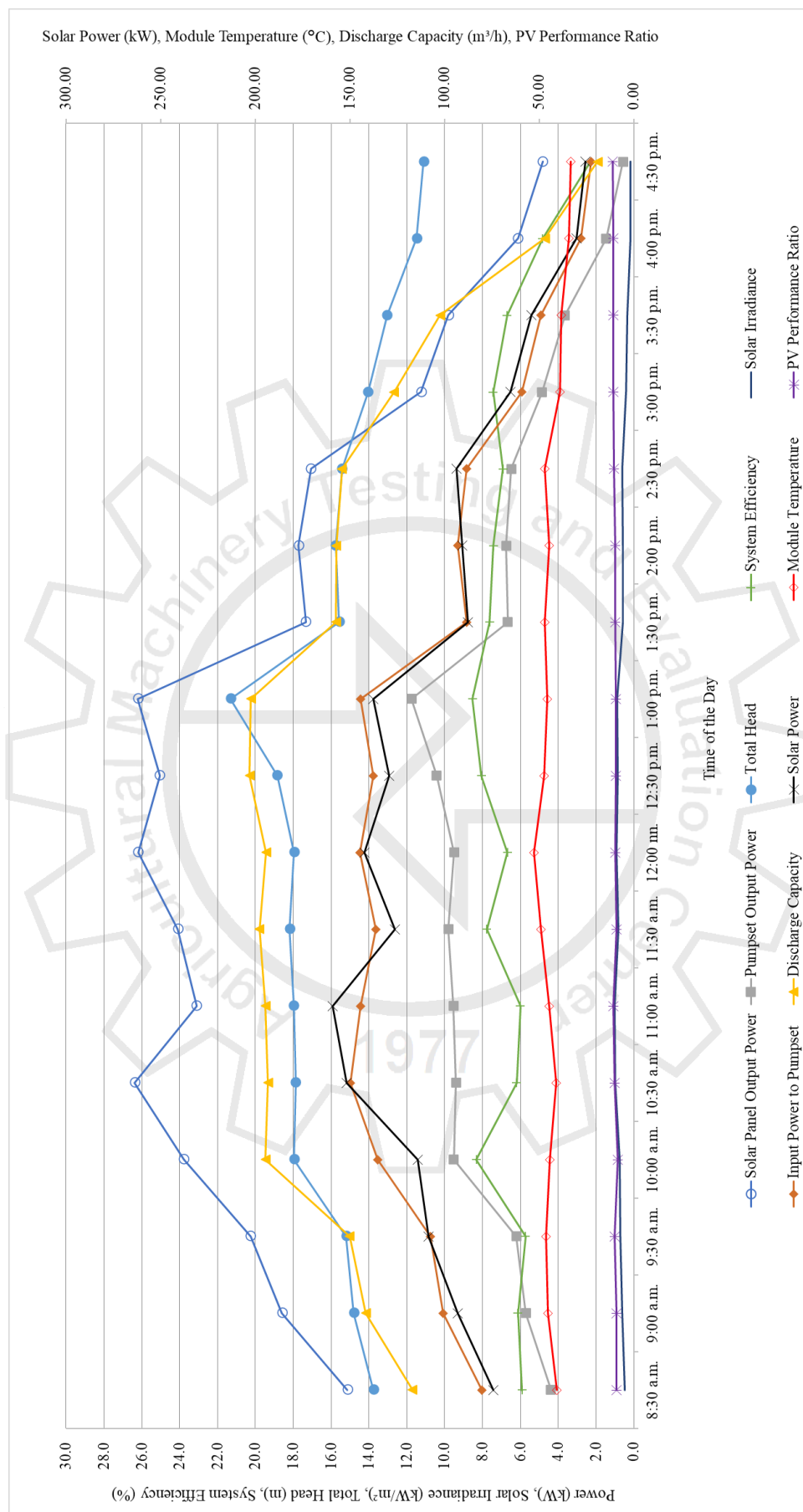
RESULTS (Cont'n)

Figure 3. Performance curves of the Solar-Powered Irrigation System installed in Brgy. Candelaria, Dagohoy, Bohol.

OBSERVATIONS

Figure 4. Nameplate of the submersible pumpset.



Figure 5. Nameplate of the controller/inverter.

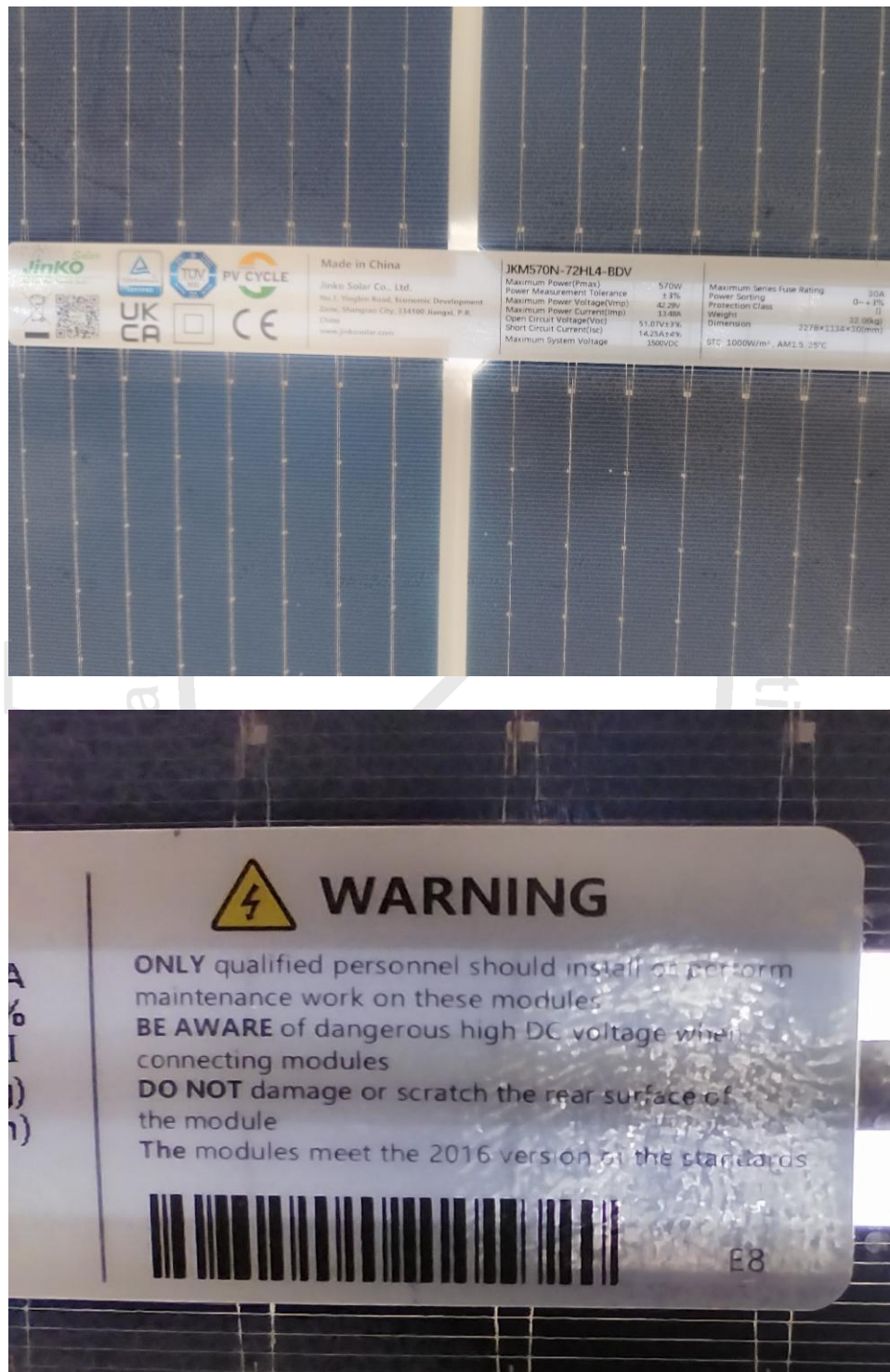
OBSERVATIONS (Cont'n)

Figure 6. Nameplate and safety notices of the PV module.

Tested by:

**FRANKLIN C. PAJARON**

Test Engineer

Verified by:

**RHEY MARC C. ALBALOS**

Test Engineer

Approved for Release:


ARTHUR L. FAJARDO
AMTEC Director

27 NOV 2024

Date

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